The Mathematics Physics Demands

A Coordinate-Free Progression from Superposition to Anyons

Paul Gustafson

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Contents

Introduction

Why does physics repeatedly demand new mathematics? A vibrating string first asks only that possible motions can be added. Quantum measurement then asks for inner products and spectral projections. Molecular spectra ask which transformations preserve a system. Electromagnetism asks how local data fit together over spacetime. Quantized Hall response asks which global features survive deformation. Open quantum systems ask that positivity remain valid after arbitrary auxiliary systems are attached. Anyons finally ask for a language in which fusion and exchange are operations rather than afterthoughts.

This book follows those demands in the order

$$\begin{gathered}\text{superposition} \longrightarrow \text{symmetry} \longrightarrow \text{fields}\\ \longrightarrow \text{topology} \longrightarrow \text{noncommutativity} \longrightarrow \text{categorification.}\end{gathered}$$

The sequence is conceptual, not historical. Physics decides why a structure enters; mathematical prerequisites decide exactly where it can enter.

Each chapter begins with one physical question. It then introduces the smallest coordinate-free language that makes the question precise, proves one central structural theorem, returns to the physical phenomenon, and ends with exercises that connect the structure to models or experiments. An experimental observation is treated as data to be predicted or organized, not as a proof of a mathematical formalism. Every model states the assumptions under which its conclusions are intended.

The companion volume *Coordinate-free Mathematics* develops a broader Algebra–Geometry–Analysis narrative. The present book is not a rearranged edition of it. It has independent numbering, transitions, proofs, exercises, and scope; the provenance ledger records the limited passages adapted from the companion.

Part I

States, Superposition, and Measurement

Chapter 1

Waves force linearity

Physical question. Why can two waves occupy the same string at the same time without destroying one another?

Model and assumptions. The string is homogeneous, perfectly flexible, held at constant tension, and displaced transversely by a small amount. Slopes are small enough that quadratic geometric corrections, damping, and external forcing may be neglected.

1.1 What language can express superposition?

A real *vector space* is a set V with addition and scalar multiplication satisfying the usual distributive, associative, and inverse laws. A map $T : V \to W$ is *linear* when

$$T(av + bw) = aT(v) + bT(w).$$

These axioms are not introduced because coordinates are convenient. They encode the empirical modeling rule that amplitudes add and may be rescaled.

For a string of length L, let $u(x,t)$ be its transverse displacement. In the small-slope model, force balance gives

$$\rho\,\partial_t^2 u = T\,\partial_x^2 u, \qquad u(0,t) = u(L,t) = 0, \tag{1.1}$$

where $T > 0$ is tension and $\rho > 0$ is linear density.

1.2 What does the wave equation force?

Theorem 1.1 (Superposition). *The solutions of* (1.1) *form a real vector space. For every fixed time t, evolution from admissible initial data* (u_0, v_0)

to $(u(\cdot,t), \partial_t u(\cdot,t))$ is linear whenever the initial-value problem has a unique solution.

Proof. Let $D = \rho\partial_t^2 - T\partial_x^2$. Differentiation is linear, so $D(au + bv) = aD(u) + bD(v)$. The boundary conditions are homogeneous and are therefore preserved by the same combination. Hence solutions are closed under addition and scaling. If E_t denotes evolution, then $aE_t(u_0, v_0) + bE_t(w_0, z_0)$ solves the equation with initial data $(au_0 + bw_0, av_0 + bz_0)$. Uniqueness identifies this solution with $E_t(au_0 + bw_0, av_0 + bz_0)$. $\square$

1.3 What does this say about an observed string?

The theorem predicts that independently prepared small-amplitude normal modes pass through one another and that their measured displacement is the sum of their displacements. It does not predict exact superposition for a large displacement: there the omitted nonlinear terms invalidate the model. For fixed endpoints the functions

$$u_n(x,t) = \sin\Big(\frac{n\pi x}{L}\Big)\cos\Big(\frac{n\pi}{L}\sqrt{T/\rho}\,t\Big)$$

are modes, and linearity makes every finite linear combination another motion. The coefficients are coordinates used to calculate a motion; the solution space and its linear evolution exist before a basis is chosen.

1.4 What further contact should be proved?

Exercise 1.1 (Energy distinguishes the physical solution space)**.** *For sufficiently regular solutions of* (1.1)*, prove conservation of*

$$E(t) = \frac{1}{2}\int_0^L \big(\rho|\partial_t u|^2 + T|\partial_x u|^2\big)\,dx.$$

Explain which boundary term uses the fixed-end assumption.

Exercise 1.2 (Critical damping breaks conservative mode evolution)**.** *Write $m\ddot{q} + b\dot{q} + kq = 0$ as a first-order linear system. Classify its three damping regimes by the characteristic polynomial and identify the nontrivial Jordan block at $b^2 = 4mk$.*

Chapter 2

Composite systems require tensor products

Physical question. If two systems are prepared independently, what state space contains both independent preparations and correlations between them?

Model and assumptions. Only the linear structure of each component is retained. The preparation rule is linear in either component when the other component is fixed.

2.1 Why are pairs of vectors not enough?

The direct sum $V \oplus W$ describes alternatives or independent sectors: a vector is a pair (v, w). A quotient V/N declares vectors differing by an element of N equivalent. Neither construction represents a preparation that is bilinear in two inputs.

The *tensor product* is a vector space $V \otimes W$ with a bilinear map $\tau : V \times W \to V \otimes W$ such that every bilinear $b : V \times W \to X$ factors through a unique linear map $\widetilde{b} : V \otimes W \to X$:

$$b = \widetilde{b} \circ \tau.$$

This universal property, rather than a choice of product basis, defines the composite state space.

2.2 What makes the tensor product canonical?

Theorem 2.1 (Existence and uniqueness of the tensor product)**.** *Every pair of vector spaces has a tensor product. Any two tensor products satisfying the universal property are related by a unique isomorphism that preserves pure tensors.*

Proof. Take the free vector space on $V \times W$ and quotient by the span of the four families enforcing additivity and homogeneity in each variable. The class of (v, w) is denoted $v \otimes w$. A bilinear map vanishes on those relations, so its linear extension descends uniquely to the quotient. If (T, τ) and (T', τ') both satisfy the property, universality gives maps $f : T \to T'$ and $g : T' \to T$ preserving pure tensors. Uniqueness applied to the identity factorization gives $gf = \mathrm{id}_T$ and $fg = \mathrm{id}_{T'}$. □

2.3 Where does entanglement enter?

A nonzero vector $\psi \in V \otimes W$ is a *product vector* if $\psi = v \otimes w$; otherwise it is entangled. The universal property permits all bilinear product preparations, but the resulting vector space necessarily contains their linear combinations. Thus entanglement is not an added force or interaction. It is already present in the minimal linear space closed under superposition of composite preparations.

2.4 What further contact should be proved?

Exercise 2.1 (Schmidt decomposition)**.** *For finite-dimensional complex inner-product spaces H_A, H_B, associate to $\psi \in H_A \otimes H_B$ the map $H_A^* \to H_B$. Apply singular-value decomposition to prove $\psi = \sum_i s_i a_i \otimes b_i$ with orthonormal families and $s_i > 0$. Prove that ψ is a product vector exactly when one Schmidt coefficient is nonzero.*

Exercise 2.2 (Universal constructions)**.** *Give coordinate-free universal properties for direct sums and quotients. Use them to prove $k(X \sqcup Y) \cong kX \oplus kY$ and $k(X \times Y) \cong kX \otimes kY$ for free vector spaces.*

Chapter 3

Measurement requires Hilbert-space geometry

Physical question. How can a state assign probabilities to mutually exclusive measurement outcomes without choosing coordinates?

Model and assumptions. The system is finite dimensional, pure states are rays in a complex vector space, and an ideal repeatable measurement is represented by a self-adjoint operator.

3.1 What structure turns amplitudes into probabilities?

A Hermitian inner product on a complex vector space H is positive definite, linear in its second argument, and conjugate linear in its first. It gives $\|\psi\|^2 = \langle\psi, \psi\rangle$. The adjoint of $A : H \to H$ is the unique A^* satisfying $\langle A\psi, \phi\rangle = \langle\psi, A^*\phi\rangle$. An operator is self-adjoint when $A = A^*$ and normal when $A^*A = AA^*$.

An orthogonal projection is an operator $P = P^* = P^2$. If $\|\psi\| = 1$, the number $\langle\psi, P\psi\rangle = \|P\psi\|^2$ lies in $[0, 1]$ and is independent of the phase chosen to represent the ray of ψ.

3.2 Why do observables have spectral outcomes?

Theorem 3.1 (Finite-dimensional spectral theorem)**.** *Every normal operator on a finite-dimensional complex inner-product space has a unique* *decomposition*

$$A = \sum_{\lambda \in \sigma(A)} \lambda P_\lambda, \qquad P_\lambda P_\mu = 0 \ (\lambda \neq \mu), \qquad \sum_\lambda P_\lambda = 1,$$

where P_λ projects onto the λ-eigenspace.

Proof. The characteristic polynomial gives an eigenvector v. Normality implies that $v^\perp$ is invariant under both A and A^*: for $w \perp v$, use $\ker(A-\lambda)^\perp = \operatorname{im}(A^* - \overline{\lambda})$. Induction on dimension produces an orthonormal eigenbasis. Grouping basis vectors by eigenvalue gives the projections. Their images are the intrinsic eigenspaces, so the decomposition is unique. □

3.3 How does the theorem return to measurement?

For a self-adjoint observable A, all eigenvalues are real. The ideal measurement model assigns

$$\Pr_\psi(A = \lambda) = \langle \psi, P_\lambda \psi \rangle, \qquad \sum_\lambda \Pr_\psi(A = \lambda) = 1.$$

The theorem supplies mutually exclusive outcomes and the inner product turns their components into normalized probabilities. This is a mathematical model of measurement statistics; agreement of repeated experiments with these frequencies tests the model rather than proving the spectral theorem.

3.4 What further contact should be proved?

Exercise 3.1 (Uncertainty from noncommutativity)**.** *For self-adjoint A, B and unit ψ, set $\Delta_\psi A = \|(A - \langle A \rangle_\psi)\psi\|$. Use Cauchy–Schwarz to prove*

$$\Delta_\psi A \, \Delta_\psi B \geq \frac{1}{2} |\langle \psi, [A, B]\psi \rangle|.$$

State precisely when equality holds.

Exercise 3.2 (Normal modes are spectral measurements)**.** *For a finite chain of equal masses and springs, express its stiffness as a positive self-adjoint operator. Find its spectral projections and recover the mode frequencies without treating the chosen mass coordinates as intrinsic.*

Chapter 4

Ancillas force complete positivity

Physical question. Why must a physical evolution remain positive after the system is coupled to an arbitrary auxiliary system on which nothing is done?

Model and assumptions. States of an n-level system are density operators in $M_n(\mathbb{C})$. Dynamics is linear on statistical mixtures, and an unused ancilla evolves by the identity map.

4.1 Why is positivity alone insufficient?

A density operator is $\rho \geq 0$ with $\operatorname{tr} \rho = 1$. A linear map $\Phi : M_n \to M_m$ is positive when it carries positive matrices to positive matrices. It is *completely positive* when $\mathrm{id}_{M_r} \otimes \Phi$ is positive for every r. A quantum channel is a completely positive trace-preserving map.

The transpose is positive but $\mathrm{id}_{M_2} \otimes t$ sends the maximally entangled two-qubit state to an operator with a negative eigenvalue. Ordinary positivity therefore fails the untouched-ancilla test.

For $\Phi : M_n \to M_m$, define the Choi matrix

$$C_\Phi = \sum_{i,j} E_{ij} \otimes \Phi(E_{ij}).$$

4.2 What structure characterizes physical finite-dimensional maps?

Theorem 4.1 (Choi–Kraus–Stinespring). *For a linear map $\Phi : M_n \to M_m$, the following are equivalent:*

1. *Φ is completely positive;*
2. *$C_\Phi \geq 0$;*
3. *$\Phi(x) = \sum_{a=1}^{r} K_a x K_a^*$ for some $K_a : \mathbb{C}^n \to \mathbb{C}^m$.*

Consequently $\Phi(x) = V^*(x \otimes 1_E)V$ for a map $V : \mathbb{C}^m \to \mathbb{C}^n \otimes E$. If Φ is unital, V is an isometry. [3, 4]

Proof. Complete positivity makes $C_\Phi = (\mathrm{id} \otimes \Phi)(|\Omega\rangle\langle\Omega|)$ positive, with $\Omega = \sum_i e_i \otimes e_i$. A positive decomposition $C_\Phi = \sum_a |v_a\rangle\langle v_a|$ identifies each v_a with a matrix K_a and direct substitution gives the Kraus formula. That formula is positive after every matrix amplification. Finally set $V\xi = \sum_a K_a^* \xi \otimes e_a$; then $V^*(x \otimes 1)V = \sum_a K_a x K_a^*$. The identity $V^*V = \Phi(1)$ proves the last assertion. □

4.3 What does dilation say about open evolution?

In the Schrödinger picture the adjoint form of the theorem says that every channel can be written

$$\rho \longmapsto \mathrm{Tr}_E(U(\rho \otimes \sigma)U^*)$$

after enlarging the environment if necessary. Irreversibility is therefore compatible with reversible evolution of a larger closed system: information is lost only when the environment is ignored. Complete positivity is the coordinate-free consistency condition that makes this statement survive every added ancilla.

4.4 What further contact should be proved?

Exercise 4.1 (No universal cloner). *Suppose a channel maps $\rho \otimes \sigma$ to $\rho \otimes \rho$ for every pure state ρ. Use monotonicity of fidelity, or contractivity of trace distance, to derive a contradiction from two nonorthogonal states.*

Exercise 4.2 (Part I synthesis). *Begin with two normal modes of the string in Chapter 1, regard their complex span as a two-level system, attach a two-level environment, and construct a unitary coupling whose reduced evolution is a dephasing channel. Find its Kraus operators and Choi matrix, prove complete positivity, and identify exactly which superpositions lose phase coherence.*

Why the next framework is necessary. Linear spaces, tensor products, and channels describe states and evolutions. They do not identify which transformations preserve the physical structure or how those transformations label observable sectors. That missing classification is symmetry.

Part II

Symmetry and Quantum Numbers

Chapter 5

Symmetry separates state space into sectors

Physical question. Why do energy levels occur in symmetry-related families rather than as unrelated individual states?

Model and assumptions. A symmetry is an invertible transformation preserving the physical predictions of the model. The transformations compose as a finite group G and act linearly on a finite-dimensional complex state space V.

5.1 How does a physical transformation become algebra?

A group representation is a homomorphism $\rho : G \to \mathrm{GL}(V)$. A subspace $W \subseteq V$ is invariant when $\rho(g)W \subseteq W$ for all g. It is irreducible when its only invariant subspaces are 0 and itself. A map $T : V \to W$ intertwines two representations when $T\rho_V(g) = \rho_W(g)T$.

If a Hamiltonian H obeys $H\rho(g) = \rho(g)H$, then every eigenspace of H is invariant. Symmetry can therefore organize degeneracy only if invariant spaces can be decomposed into elementary pieces.

5.2 Why can finite symmetry be completely decomposed?

Theorem 5.1 (Maschke and Schur)**.** *Every finite-dimensional complex representation of a finite group is a direct sum of irreducible representations. If*

V, W are irreducible, every nonzero intertwiner $V \to W$ is an isomorphism; moreover $\mathrm{End}_G(V) = \mathbb{C}\, 1_V$*.*

Proof. Average any Hermitian inner product over G to make it invariant. If $W \subseteq V$ is invariant, then $W^\perp$ is invariant as well. Induction on dimension gives a decomposition into irreducibles. For an intertwiner T, its kernel and image are invariant, so irreducibility makes a nonzero T invertible. If $T \in \mathrm{End}_G(V)$, choose an eigenvalue λ over $\mathbb{C}$. The intertwiner $T - \lambda 1$ has nonzero kernel and is therefore zero. $\square$

5.3 What does decomposition predict physically?

Write

$$V \cong \bigoplus_\alpha M_\alpha \otimes V_\alpha,$$

where the V_α are inequivalent irreducibles and G acts trivially on the multiplicity spaces M_α. Every symmetry-preserving Hamiltonian has the form $\bigoplus_\alpha H_\alpha \otimes 1_{V_\alpha}$. Thus an energy inside the α sector has at least $\dim V_\alpha$-fold degeneracy unless other structure splits the multiplicity space. The claim is conditional on the symmetry being exact; perturbations that break G may split the level.

5.4 What further contact should be proved?

Exercise 5.1 (Averaging produces invariant projectors)**.** *For $W \subseteq V$ invariant and any projection $P : V \to W$, prove that*

$$\widetilde{P} = \frac{1}{|G|} \sum_{g \in G} \rho(g) P \rho(g)^{-1}$$

is an equivariant projection onto W.

Exercise 5.2 (Symmetry protects degeneracy)**.** *Let an irreducible representation V_α occur once in an eigenspace of a G-invariant Hamiltonian. Prove that every G-invariant perturbation acts there by a scalar. Explain why this conclusion says nothing about a perturbation that breaks G.*

Chapter 6

Characters predict molecular spectra

Physical question. Why are some molecular vibrations visible in infrared or Raman spectra while symmetry forbids others?

Model and assumptions. The equilibrium molecule has a finite point group G. Small nuclear displacements form a linear representation, and the electric dipole approximation is valid for the transitions under discussion.

6.1 Which invariant detects an irreducible sector?

The character of a representation is $\chi_V(g) = \mathrm{tr}(\rho_V(g))$. It is constant on conjugacy classes and does not depend on a basis. For class functions define

$$\langle \chi, \psi \rangle_G = \frac{1}{|G|} \sum_{g \in G} \overline{\chi(g)} \psi(g).$$

For an irreducible V_α of dimension d_α, the operator

$$P_\alpha = \frac{d_\alpha}{|G|} \sum_{g \in G} \overline{\chi_\alpha(g)} \rho_V(g)$$

is the candidate projector onto its isotypic component.

6.2 Why do characters recover multiplicities?

Theorem 6.1 (Character orthogonality)**.** *Irreducible characters of a finite group are orthonormal. If* $V \cong \bigoplus_\alpha m_\alpha V_\alpha$*, then*

$$m_\alpha = \langle \chi_\alpha, \chi_V \rangle_G,$$

and P_α *is the projection onto the* V_α*-isotypic component.*

Proof. For a linear map $A : V_\alpha \to V_\beta$, average $A \mapsto |G|^{-1} \sum_g \rho_\beta(g) A \rho_\alpha(g)^{-1}$. The result is an intertwiner. Schur's lemma from Chapter 5 makes it zero for $\alpha \neq \beta$ and a scalar identity for $\alpha = \beta$. Applying this to matrix units and taking traces gives character orthogonality. Additivity of trace on direct sums then gives the multiplicity formula. Substitution into P_α shows that it is identity on copies of V_α and zero on all other irreducibles. □

6.3 How does this return to a water spectrum?

For water at equilibrium the model point group is C_{2v}. The nuclear displacement representation has dimension nine. Removing translations and rotations leaves

$$V_{\text{vib}} \cong 2A_1 \oplus B_2.$$

A dipole transition from V_i to V_f through a dipole component V_μ can be nonzero only if

$$\mathrm{Hom}_G(1, V_f^* \otimes V_\mu \otimes V_i) \neq 0.$$

This invariant-tensor condition predicts allowed lines. Measured line positions additionally depend on masses, force constants, anharmonicity, and instrumental conditions; symmetry organizes visibility but does not determine the frequencies by itself. Reference vibrational frequencies provide the experimental comparison for this classification [5].

6.4 What further contact should be proved?

Exercise 6.1 (Water has three symmetry-classified vibrations)**.** *Construct the* C_{2v} *character of the nine-dimensional displacement space from the fixed nuclei and spatial traces. Subtract translations and rotations and derive* $2A_1 \oplus B_2$.

Exercise 6.2 (Raman selection differs from dipole selection)**.** *Let the polarizability transform as* $\mathrm{Sym}^2(V_\mu)$*. Determine which* C_{2v} *irreducibles occur and compare the Raman-active and infrared-active water modes.*

Chapter 7

Continuous symmetry produces angular momentum

Physical question. Why are rotational quantum numbers discrete even though physical rotations vary continuously?

Model and assumptions. Rotations act continuously and unitarily on a finite-dimensional state space. Infinitesimal generators are defined on the whole space; domain issues of unbounded operators are postponed.

7.1 How is continuous transformation differentiated?

A matrix Lie group is a closed subgroup of $\mathrm{GL}(n, \mathbb{C})$. Its Lie algebra consists of the matrices X for which e^{tX} remains in the group, with bracket $[X, Y] = XY - YX$. Differentiating a smooth representation $U : G \to U(H)$ gives a Lie-algebra representation. For rotations, define self-adjoint generators J_i by differentiating the three one-parameter rotation subgroups; with $\hbar = 1$ they satisfy

$$[J_i, J_j] = i\varepsilon_{ijk}J_k.$$

Set $J^2 = J_1^2 + J_2^2 + J_3^2$.

7.2 Which operator labels rotational sectors?

Theorem 7.1 (The rotational Casimir)**.** *The operator J^2 commutes with every J_i and hence with the connected rotation representation. On every* *irreducible complex rotation sector it is a scalar.*

Proof. Using $[AB, C] = A[B, C] + [A, C]B$,

$$[J^2, J_\ell] = i \sum_{i,k} \varepsilon_{i\ell k}(J_i J_k + J_k J_i) = 0,$$

because the coefficient is antisymmetric in i, k while the parenthesis is symmetric. Thus J^2 intertwines the Lie-algebra action and therefore the connected group action. Schur's lemma makes it scalar on an irreducible sector. $\square$

7.3 Why do angular-momentum quantum numbers result?

Introduce $J_\pm = J_1 \pm iJ_2$. The commutators $[J_3, J_\pm] = \pm J_\pm$ make $J_\pm$ raising and lowering operators. In a finite-dimensional unitary irreducible sector, positivity of $J_\mp J_\pm$ forces a highest weight $j \in \frac{1}{2}\mathbb{Z}_{\geq 0}$ and weights $m = -j, -j+1, \ldots, j$, while $J^2 = j(j+1)$. Integer j descends to an honest representation of $SO(3)$; half-integer j is an honest representation of its double cover $SU(2)$ and only a projective representation of $SO(3)$. The continuous group therefore produces discrete labels through unitarity and finite dimensionality.

7.4 What further contact should be proved?

Exercise 7.1 (Spin one-half)**.** *Set $J_i = \sigma_i/2$ for the Pauli matrices. Verify the commutation relations, compute J^2, and prove that a 2π rotation acts by -1 while a 4π rotation acts by 1.*

Exercise 7.2 (Rotational invariance conserves angular momentum)**.** *Let canonical position and momentum operators satisfy $[Q_i, P_j] = i\delta_{ij}$ and set $L_i = \sum_{jk} \varepsilon_{ijk} Q_j P_k$. For $H = (P_1^2 + P_2^2 + P_3^2)/(2m) + V(Q_1^2 + Q_2^2 + Q_3^2)$, prove $[H, L_i] = 0$ and identify precisely where rotational invariance of the potential is used.*

Chapter 8

Quantum phases require projective symmetry

Physical question. How can a symmetry act consistently on physical rays even when its operators fail to multiply exactly as the group does?

Model and assumptions. State vectors differing by a phase represent the same pure state. A discrete symmetry group acts unitarily on rays, and phases belong to $U(1)$ with trivial group action.

8.1 Which phases are removable?

A projective representation has operators satisfying

$$U_g U_h = \alpha(g,h) U_{gh}, \qquad \alpha(g,h) \in U(1).$$

Associativity forces the cocycle equation

$$\alpha(g,h)\alpha(gh,k) = \alpha(h,k)\alpha(g,hk).$$

Rephasing U_g by $\lambda(g)$ changes α by the coboundary $\lambda(g)\lambda(h)\lambda(gh)^{-1}$. The quotient of cocycles by coboundaries is $H^2(G, U(1))$.

8.2 Why does cohomology classify projective actions?

Theorem 8.1 (Projective classification)**.** *Equivalence classes of normalized factor systems for projective unitary representations of G are classified by $H^2(G, U(1))$. Tensor product multiplies the corresponding cohomology classes.*

Proof. The displayed associativity calculation says exactly that a factor system is a normalized 2-cocycle. Changing representatives of rays rephases the operators and hence multiplies the cocycle by a coboundary; conversely every coboundary is produced by such a rephasing. For tensor products, $(U_g \otimes V_g)(U_h \otimes V_h) = \alpha(g,h)\beta(g,h)U_{gh} \otimes V_{gh}$, so classes multiply. □

8.3 How can a cohomology class protect an edge state?

Let $G = \mathbb{Z}/2 \times \mathbb{Z}/2$ be generated by two π rotations. On a spin-half edge take $U_x = \sigma_1$ and $U_z = \sigma_3$. Their anticommutation realizes the nontrivial element of $H^2(G, U(1)) \cong \mathbb{Z}/2$. Any local edge operator commuting with both is scalar, so a symmetry-preserving local perturbation cannot split the doublet. In the spin-one Haldane model, the statement relies on a bulk gap, locality, and preservation of the specified symmetry; closing the gap or breaking the symmetry evades the conclusion.

8.4 What further contact should be proved?

Exercise 8.1 (The projective class squares to zero)**.** *Compute the factor system for U_x, U_z above and prove that its tensor square is cohomologically trivial. Interpret this as the possibility of combining two protected edges into an ordinary linear representation.*

Exercise 8.2 (Part II synthesis)**.** *Let a Hamiltonian commute with a finite symmetry group and suppose one isotypic sector carries a projective edge action. Use characters to isolate the sector, Schur's lemma to constrain symmetric perturbations, and $H^2(G, U(1))$ to decide whether the action can be rephased to an ordinary representation. Separate degeneracy forced by an irreducible dimension from degeneracy protected by a nontrivial projective class.*

Why the next framework is necessary. Global symmetry uses the same group at every point. A field must compare locally varying values and frames across spacetime. Making symmetry local forces differential forms, metrics, bundles, connections, and curvature.

Part III

Classical Fields and Geometry

Chapter 9

Maxwell fields are differential forms

Physical question. How can electric charge and magnetic flux be described in a way that every observer can integrate consistently?

Model and assumptions. Spacetime is a smooth oriented four-manifold. The electromagnetic field is classical, sources are smooth on the region under study, and singular charges or material interfaces are excluded unless they are represented by removed subsets or distributions.

9.1 What can be pulled back and integrated?

A smooth n-manifold is a Hausdorff, second-countable space locally modeled on $\mathbb{R}^n$ by smoothly compatible charts. Its tangent and cotangent bundles are TM and T^*M. A differential p-form is a smooth section of $\Lambda^p T^*M$; write $\Omega^p(M)$ for their space.

The wedge product is alternating and the exterior derivative $d : \Omega^p(M) \to \Omega^{p+1}(M)$ is the unique graded derivation extending the differential of functions and satisfying $d^2 = 0$. For a smooth map $f : N \to M$, pullback obeys $f^*d = df^*$. Integration pairs compactly supported top-degree forms with oriented manifolds. Stokes' theorem states

$$\int_X d\omega = \int_{\partial X} \omega.$$

Closed forms lie in $\ker d$, exact forms in $\operatorname{im} d$, and $H^p_{\mathrm{dR}}(M) = \ker d / \operatorname{im} d$ records the obstruction to finding a global potential.

9.2 Which conservation law is structural?

Theorem 9.1 (Maxwell charge conservation). *Let $F \in \Omega^2(M)$ be the electromagnetic field and let $J \in \Omega^3(M)$ be the current. If a metric-dependent Hodge star satisfies the Maxwell equations*

$$dF = 0, \qquad d{\star}F = J,$$

then $dJ = 0$. For every compact oriented four-region X,

$$\int_{\partial X} J = 0.$$

Proof. Apply d to the sourced equation: $dJ = d^2{\star}F = 0$. Stokes' theorem then gives $\int_{\partial X} J = \int_X dJ = 0$. □

9.3 How does this return to charge measurements?

The boundary of a spacetime slab consists of two spatial slices and a timelike side. The theorem says that the difference of total charge on the slices equals the current flux through the side. This conclusion does not depend on a coordinate continuity equation; that equation is a chart expression of $dJ = 0$. The physical model still assumes that J includes every relevant source. Apparent nonconservation may instead signal current crossing an unmodeled boundary or failure of the classical description.

9.4 What further contact should be proved?

Exercise 9.1 (Flux depends only on homology). *If F is closed and two oriented 2-cycles differ by the boundary of a 3-chain, prove that their integrals of F agree. State the dual dependence on the de Rham class $[F]$.*

Exercise 9.2 (Gauge potentials are local data). *If $F = dA$ on a contractible region, prove that replacing A by $A + d\lambda$ does not change F. Give an example on a punctured region where a closed one-form is not exact.*

Chapter 10

Metrics turn geometry into dynamics

Physical question. What additional structure turns invariant field forms into lengths, wave equations, freely falling paths, and tidal acceleration?

Model and assumptions. The manifold carries a smooth nondegenerate metric of fixed signature. Test bodies are small enough to neglect their backreaction and, when geodesic motion is invoked, non-gravitational forces are absent.

10.1 What does a metric determine without coordinates?

A pseudo-Riemannian metric is a smooth nondegenerate symmetric bilinear form g on each tangent space. An orientation and g determine a volume form vol_g and a Hodge star by

$$\alpha \wedge \star\beta = \langle \alpha, \beta \rangle_g \, \mathrm{vol}_g \, .$$

A connection on TM assigns covariant derivatives $\nabla_X Y$, is linear in X, satisfies a Leibniz rule in Y, and permits vectors at nearby points to be compared infinitesimally. It is torsion free when $\nabla_X Y - \nabla_Y X = [X, Y]$ and metric compatible when $\nabla g = 0$.

10.2 Why is there a canonical metric connection?

Theorem 10.1 (Levi–Civita)**.** *Every pseudo-Riemannian metric has a unique torsion-free, metric-compatible connection. It is determined by the coordinate-free Koszul formula*

$$\begin{aligned} 2g(\nabla_X Y, Z) &= Xg(Y,Z) + Yg(Z,X) - Zg(X,Y) \\ &\qquad - g(X,[Y,Z]) + g(Y,[Z,X]) + g(Z,[X,Y]). \end{aligned}$$

Proof. Metric compatibility expands each derivative of an inner product into two covariant-derivative terms. Add the expansions for $Xg(Y,Z)$ and $Yg(Z,X)$, subtract the one for $Zg(X,Y)$, and use zero torsion to replace differences of covariant derivatives by Lie brackets. This yields the formula and hence uniqueness by nondegeneracy of g. Conversely the formula defines $\nabla_X Y$ uniquely; direct substitution verifies linearity, the Leibniz rule, zero torsion, and metric compatibility. □

10.3 How do connection and star return to physics?

The Hodge star in Chapter 9 supplies the metric part of Maxwell's equations. The Levi–Civita connection defines geodesics by $\nabla_{\dot\gamma}\dot\gamma = 0$ and curvature by

$$R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z.$$

For a variation of geodesics with separation field J, the equation $\nabla_T \nabla_T J = R(T,J)T$ turns curvature into relative acceleration. A gravitational-wave interferometer uses this tidal effect; it does not measure coordinate components of the metric in isolation.

10.4 What further contact should be proved?

Exercise 10.1 (The Hodge star is unique)**.** *Prove pointwise existence and uniqueness of $\star$ from the induced inner product on exterior powers. Determine $\star^2$ in terms of form degree and metric signature.*

Exercise 10.2 (Killing fields give conserved quantities)**.** *If X satisfies $\mathcal{L}_X g = 0$ and γ is an affinely parametrized geodesic, prove that $g(X,\dot\gamma)$ is constant.*

Chapter 11

Hamiltonian evolution is symplectic

Physical question. Why does conservative time evolution preserve phase-space volume even when it stretches and folds individual regions?

Model and assumptions. The classical system has a finite-dimensional smooth phase space, its dynamics is deterministic, and the Hamiltonian has no explicit time dependence. Dissipation and stochastic forcing are excluded.

11.1 What replaces a metric on phase space?

A symplectic form is a closed nondegenerate two-form ω on a manifold M. Nondegeneracy identifies every one-form with a unique vector field. A Hamiltonian $H \in C^\infty(M)$ therefore determines X_H by

$$\iota_{X_H}\omega = dH.$$

The Poisson bracket is $\{f, h\} = \omega(X_f, X_h)$. In canonical coordinates the definition reproduces Hamilton's equations, but neither ω nor X_H depends on those coordinates.

11.2 Why can Hamiltonian flow not compress phase volume?

Theorem 11.1 (Liouville)**.** *The local flow φ_t of a Hamiltonian vector field preserves the symplectic form and the Liouville volume $\omega^n/n!$ on a $2n$-dimensional phase space.*

Proof. Cartan's formula and $d\omega = 0$ give

$$\mathcal{L}_{X_H}\omega = d(\iota_{X_H}\omega) + \iota_{X_H}d\omega = d^2H = 0.$$

Thus $\frac{d}{dt}\varphi_t^*\omega = \varphi_t^*\mathcal{L}_{X_H}\omega = 0$, so $\varphi_t^*\omega = \omega$. Pullback respects wedge products, hence it also fixes $\omega^n/n!$. □

11.3 What physical consequence follows?

An ensemble transported by a Hamiltonian flow cannot converge into a smaller phase-volume attractor. This explains why ordinary friction cannot be represented by an autonomous Hamiltonian on the same finite-dimensional phase space: friction contracts volumes and violates the theorem's assumptions. In equilibrium statistical mechanics the preserved volume is the invariant measure from which microcanonical counting begins.

11.4 What further contact should be proved?

Exercise 11.1 (Magnetic curvature gives the Lorentz force)**.** *On T^*Q replace the canonical symplectic form by $\omega_B = \omega_0 + q\pi^*B$ for a closed two-form B on Q. For kinetic Hamiltonian H, derive the Lorentz-force equation.*

Exercise 11.2 (Moment maps recover angular momentum)**.** *For the lifted rotation action on $T^*\mathbb{R}^3$, prove that the moment map is $J(q,p) = q \times p$. Show directly that a rotation-invariant Hamiltonian conserves J, and compare with Chapter 7.*

Chapter 12

Local symmetry becomes a connection

Physical question. How can internal states at different space-time points be compared when no preferred frame exists globally?

Model and assumptions. Internal states form fibers of a smooth vector bundle. A compact Lie group G acts on the fibers, local gauge transformations are changes of frame, and classical gauge fields are smooth connections.

12.1 Which object compares neighboring fibers?

A vector bundle $E \to M$ is locally a product $U \times V$ with linear transition maps. A connection is a linear map

$$\nabla : \Omega^0(M, E) \to \Omega^1(M, E), \qquad \nabla(fs) = df \otimes s + f\nabla s.$$

Extending it to E-valued forms defines curvature $F_\nabla = \nabla^2 \in \Omega^2(M, \operatorname{End} E)$.

For a principal G-bundle, a local frame represents the connection by a Lie-algebra-valued one-form A, with

$$F_A = dA + A \wedge A.$$

A gauge transformation changes A but conjugates F_A; curvature is the local obstruction to path-independent transport.

12.2 Which identity constrains every gauge field?

Theorem 12.1 (Bianchi identity)**.** *Every connection satisfies $d_A F_A = 0$, or intrinsically $\nabla F_\nabla = 0$.*

Proof. In a local frame, $d_A\eta = d\eta + A\wedge\eta - (-1)^p\eta\wedge A$ for a p-form. Substitute $F_A = dA + A\wedge A$, use $d^2 = 0$ and the graded Leibniz rule, and observe that all remaining terms cancel associatively. Since the identity transforms covariantly, it is independent of the chosen frame. □

12.3 How does local symmetry produce field equations?

With an invariant inner product on the Lie algebra and the Hodge star from Chapter 10, the Yang–Mills action is

$$S(A) = \frac{1}{2}\int_M \langle F_A \wedge \star F_A\rangle.$$

A compactly supported variation gives $d_A \star F_A = 0$ in vacuum. The Bianchi identity is kinematic; the Yang–Mills equation is dynamical. For $G = U(1)$ they reduce to the two Maxwell equations. Nonabelian groups additionally self-couple through $A\wedge A$.

The Standard Model uses a particular gauge group and matter representations. Those choices are empirical inputs to the model; gauge geometry organizes their charges and couplings but does not derive the observed representation list from differential geometry alone.

12.4 What further contact should be proved?

Exercise 12.1 (Gauge variation of curvature)**.** *For $A^g = g^{-1}Ag + g^{-1}dg$, prove $F_{A^g} = g^{-1}F_A g$ and show that the Yang–Mills action is gauge invariant.*

Exercise 12.2 (Part III synthesis)**.** *On a principal $U(1)$-bundle, connect the four descriptions*

$$\textit{potential} \longrightarrow \textit{connection} \longrightarrow \textit{curvature} \longrightarrow \textit{Maxwell field.}$$

Derive both Maxwell equations, distinguish the Bianchi identity from the Euler–Lagrange equation, and identify exactly where orientation, metric, and bundle topology enter.

Why the next framework is necessary. Curvature and differential equations are local. They cannot detect every effect of a flat connection around a hole, nor explain why a transport coefficient remains an integer under continuous deformation. Global observables force holonomy, characteristic classes, K-theory, and index.

Part IV

Topological Quantum Phenomena

Chapter 13

A field-free loop can carry phase

Physical question. How can an electron acquire a measurable phase while traveling only through a region where the electromagnetic field vanishes?

Model and assumptions. An ideal solenoid confines magnetic flux to an inaccessible region. Outside it the particle is coherent, nonrelativistic, and described by minimal $U(1)$ coupling. Leakage fields and environmental decoherence are neglected in the ideal model.

13.1 What does the interference observation demand?

Two electron paths passing on opposite sides of a shielded solenoid recombine. The model predicts a relative phase even though $F = dA = 0$ along each path. The gauge-invariant datum is not the value of A at a point but the holonomy

$$\mathrm{Hol}_A(\gamma) = \exp\left(\frac{iq}{\hbar}\int_\gamma A\right)$$

around the closed loop obtained from the two paths.

In the field-free region M, the potential is a closed one-form. A gauge change $A \mapsto A + d\lambda$ changes its representative but not its de Rham class $[A] \in H^1_{\mathrm{dR}}(M)$.

13.2 Why is holonomy topological in the flat region?

Theorem 13.1 (Cohomology–homology dependence)**.** *For a closed one-form A and a closed loop γ, the integral $\int_\gamma A$ depends only on $[A] \in H^1_{\mathrm{dR}}(M)$ and $[\gamma] \in H_1(M;\mathbb{R})$. Consequently the Aharonov–Bohm phase is invariant under gauge change and homology-preserving deformation of the path.*

Proof. Replacing A by $A + d\lambda$ changes the integral by $\int_\gamma d\lambda = 0$. If $\gamma - \gamma' = \partial C$, then Stokes' theorem from Chapter 9 gives

$$\int_\gamma A - \int_{\gamma'} A = \int_C dA = 0.$$

Both dependencies therefore descend to the stated classes. □

13.3 What phase does the ideal solenoid predict?

For $M = \mathbb{R}^3$ minus the solenoid axis, let

$$A = \frac{\Phi}{2\pi} d\theta.$$

A once-winding loop has $\int_\gamma A = \Phi$, so the relative phase is $q\Phi/\hbar$ modulo 2π. Shielded-flux electron interferometry measures the displacement of fringes as flux varies. The experiment tests this holonomy prediction while separately controlling leakage fields; it does not prove de Rham theory [1, 2].

13.4 What further contact should be proved?

Exercise 13.1 (Winding computes holonomy)**.** *For a loop of winding number n around the removed axis, prove $\int_\gamma d\theta = 2\pi n$ and compute its phase.*

Exercise 13.2 (Flux periodicity)**.** *Show that all interference probabilities are unchanged when $q\Phi/\hbar$ changes by $2\pi k$. Identify the corresponding flux quantum.*

Chapter 14

Curvature produces stable characteristic classes

Physical question. Which global quantity detects a twisted family of quantum states even though every small patch admits a smooth frame?

Model and assumptions. The parameter space is a compact smooth manifold and the relevant eigenspaces have constant finite rank. Connections are smooth and the structure group is unitary.

14.1 How can local curvature define a global class?

For a complex vector bundle $E \to X$ with connection ∇, let F_∇ be its curvature. The total Chern form is defined by

$$\det\left(1 + \frac{i}{2\pi} F_\nabla\right) = 1 + c_1(E,\nabla) + c_2(E,\nabla) + \cdots .$$

The forms are assembled from invariant polynomials, so they agree under changes of local frame.

Stable equivalence declares E and F equivalent when $E \oplus \underline{\mathbb{C}}^r \cong F \oplus \underline{\mathbb{C}}^s$ after balancing ranks. The Grothendieck group of bundles under direct sum is $K^0(X)$.

14.2 Why do Chern classes ignore the chosen connection?

Theorem 14.1 (Chern–Weil independence). *Each Chern form is closed, and its de Rham class is independent of the connection. Direct sum makes the Chern character*

$$\mathrm{ch}(E) = \left[\mathrm{tr}\exp\left(\frac{iF_\nabla}{2\pi}\right)\right]$$

descend to a homomorphism $\mathrm{ch} : K^0(X) \to H^{\mathrm{even}}_{\mathrm{dR}}(X)$.

Proof. The Bianchi identity from Chapter 12 and invariance of trace imply $d\ \mathrm{tr}(F_\nabla^k) = 0$. For a path of connections ∇_t with derivative a_t, differentiating under the trace gives

$$\frac{d}{dt}\,\mathrm{tr}(F_t^k) = k\,d\ \mathrm{tr}(a_t \wedge F_t^{k-1}).$$

Integrating in t shows that two choices differ by an exact form. The exponential trace is additive on direct sums, so it respects the Grothendieck relation and descends to $K^0(X)$. □

14.3 What invariant results for a surface?

For a complex bundle over a closed oriented surface,

$$C_1(E) = \frac{i}{2\pi}\int_X \mathrm{tr}(F_\nabla)$$

is an integer. It is unchanged by a smooth deformation of the connection or by adding trivial bundles. Curvature may shift from place to place, but its total Chern number cannot vary continuously. The integer can jump only when the underlying bundle ceases to vary continuously within the assumed class.

14.4 What further contact should be proved?

Exercise 14.1 (A nonzero Chern number obstructs a global frame). *Prove that a bundle admitting a global frame is trivial and has vanishing positive-degree Chern classes. Conclude that* $C_1(E) \neq 0$ *obstructs a global smooth eigenbasis.*

Exercise 14.2 (Stable equivalence preserves the Chern character)**.** *Show directly that adding a trivial bundle changes only the degree-zero rank term of the Chern character.*

Chapter 15

The Hall integer is a bundle invariant

Physical question. Why does Hall conductance remain pinned to integer multiples of e^2/h while microscopic Hamiltonian parameters vary?

Model and assumptions. The crystal is noninteracting and translation invariant, the Fermi energy lies in a spectral gap, temperature is negligible compared with that gap, and the occupied bands vary smoothly over the Brillouin torus. The TKNN linear-response formula is taken as the physical bridge from the model to conductance.

15.1 How does a gapped Hamiltonian produce topology?

Let $H : k \mapsto H(k)$ be a smooth family of finite-dimensional self-adjoint operators over the Brillouin torus $\mathbb{T}^2$. A uniform gap at zero gives the occupied spectral projection

$$P(k) = \frac{1}{2\pi i} \int_{\Gamma} (z - H(k))^{-1} \, dz,$$

where Γ surrounds the negative spectrum. The images of $P(k)$ form the Bloch bundle $E_{\mathrm{occ}} \to \mathbb{T}^2$. Adding inert filled or empty flat bands changes the bundle by a trivial summand, so the stable phase is a K-theory class.

15.2 Why can the Hall integer change only at a gap closing?

Theorem 15.1 (Gap-preserving invariance)**.** *A smooth gap-preserving homotopy $H_s(k)$ gives isomorphic endpoint occupied bundles. Hence*

$$\mathrm{Ch}_1(P) = \frac{i}{2\pi} \int_{\mathbb{T}^2} \mathrm{tr}\big(P\, dP \wedge dP\big)$$

is an integer independent of s and of added trivial bands.

Proof. Uniform compactness of $[0,1] \times \mathbb{T}^2$ keeps a common contour inside the gap, so the Riesz formula produces a smooth projection on the product. Its image is a vector bundle whose restrictions at $s = 0$ and $s = 1$ are isomorphic because an interval is contractible. The displayed form represents the first Chern class by Chapter 14; its integral is therefore constant and integral. A trivial summand has zero curvature and does not change the pairing. □

15.3 How does topology return to conductance?

The TKNN formula predicts

$$\sigma_{xy} = \mathrm{Ch}_1(P)\frac{e^2}{h}.$$

This is the physical bridge from the bundle invariant to the observed response [6, 7]. Thus continuous changes of hopping amplitudes cannot move the ideal conductance away from its plateau while the gap and model assumptions remain valid. A jump requires a point where the occupied projection is no longer defined continuously—typically a gap closing. Disorder and interactions require broader formulations, but the invariant-versus-gap logic survives in operator-algebraic versions.

15.4 What further contact should be proved?

Exercise 15.1 (Berry curvature equals projection curvature)**.** *For an isolated occupied line with local unit vector $\psi(k)$, compare $i\langle\psi, d\psi\rangle$ with the connection induced by P. Prove that their curvatures yield the projection formula above.*

Exercise 15.2 (A two-band transition)**.** *For $H(k) = d(k) \cdot \sigma$ with $d(k) \neq 0$, show that the occupied Chern number is the degree of $d/|d| : \mathbb{T}^2 \to S^2$. Identify why the degree can change only where $d = 0$.*

Chapter 16

Fredholm index counts protected zero modes

Physical question. Why is the imbalance of two kinds of zero modes stable even though each kernel dimension can jump under perturbation?

Model and assumptions. Operators act between complex Hilbert spaces, have closed range, and have finite-dimensional kernel and cokernel. Perturbations are bounded and small in norm, or compact when explicitly stated.

16.1 Which difference survives perturbation?

An operator $T : H_0 \to H_1$ is Fredholm when its kernel and cokernel are finite dimensional and its range is closed. Its index is

$$\operatorname{ind} T = \dim \ker T - \dim \operatorname{coker} T.$$

The compact operators $\mathcal{K}(H)$ form an ideal in the bounded operators, and the Calkin algebra is $\mathcal{Q}(H) = \mathcal{B}(H)/\mathcal{K}(H)$.

16.2 Why is the index topologically stable?

Theorem 16.1 (Atkinson stability)**.** *T is Fredholm exactly when its image in the Calkin algebra is invertible. The Fredholm index is locally constant and is unchanged by compact perturbations.*

Proof. If T is Fredholm, choose inverses on complements of its finite-dimensional kernel and cokernel; extending by zero gives a parametrix S for which $ST - 1$ and $TS - 1$ have finite rank. Conversely a parametrix modulo compacts implies finite-dimensional kernel and cokernel and closed range. Thus Fredholm operators are the inverse image of the invertible group of $\mathcal{Q}(H)$, an open set. Along a sufficiently small norm path the kernel-cokernel difference cannot change, as follows from the parametrix and finite-dimensional reduction. Compact perturbations have the same Calkin image and can be joined through $T + sK$, so the index is unchanged. □

16.3 How does index meet geometry and physics?

An elliptic differential operator on a closed manifold has a Fredholm extension. The Atiyah–Singer theorem identifies its analytic index with a topological K-theory pairing of its principal symbol [8]. In gauge backgrounds, this converts characteristic charge into chiral zero-mode imbalance. The deep theorem is used here as a bridge; the stability proved above already explains why admissible perturbations cannot change the imbalance.

For the circle operator $D = d/d\theta$ on periodic complex functions, kernel and cokernel are both the constants, so $\operatorname{ind} D = 0$, agreeing with the Euler characteristic of S^1. This elementary case displays the analytic-topological equality without claiming a proof of the general index theorem.

16.4 What further contact should be proved?

Exercise 16.1 (The unilateral shift has index minus one)**.** *On $\ell^2(\mathbb{N})$ let $S(e_n) = e_{n+1}$. Compute the kernels and cokernels of S and S^*, then verify index stability after a finite-rank perturbation.*

Exercise 16.2 (Part IV synthesis)**.** *For the Aharonov–Bohm phase, Hall integer, and Fredholm index, identify the structured input, the deformation class, the numerical pairing, and the hypothesis whose failure permits a jump. Express the first as a cohomology–homology pairing, the second as a Chern pairing of a K-class, and the third as the boundary invariant of an invertible Calkin class.*

> **Why the next framework is necessary.** Topology explains quantized sectors but not how maps behave after arbitrary matrix amplification, how nonlocal correlations are bounded, or how

infinite observable algebras contain physical weak limits. Those questions require noncommutative analysis.

Part V

Quantum Information and Noncommutative Analysis

Chapter 17

Composite tests require matrix norms

Physical question. Why can two linear maps with the same ordinary norm behave differently when an entangled auxiliary system is attached?

Model and assumptions. Systems are finite dimensional in the explicit calculations. A subspace of operators inherits the norm arising from its action on a Hilbert space, and an r-level ancilla is represented by matrix amplification.

17.1 Which norm remembers every ancilla?

A concrete operator space is a closed subspace $E \subseteq \mathcal{B}(H, K)$. Its rth matrix level is

$$M_r(E) \subseteq \mathcal{B}(H^{\oplus r}, K^{\oplus r}).$$

For $\phi : E \to F$, set $\phi_r([x_{ij}]) = [\phi(x_{ij})]$ and

$$\|\phi\|_{\mathrm{cb}} = \sup_r \|\phi_r\|.$$

The row and column operator spaces are $R_n = M_{1,n}$ and $C_n = M_{n,1}$ with their concrete matrix norms. As Hilbert spaces both are $\mathbb{C}^n$; as operator spaces they encode different directions of action.

17.2 Can ordinary norm miss an arbitrarily large effect?

Theorem 17.1 (Transposition detects matrix levels)**.** *For transposition $t : M_n \to M_n$,*

$$\|t\| = 1, \qquad \|t\|_{\mathrm{cb}} = n.$$

Thus ordinary boundedness does not control behavior under ancillas.

Proof. Transposition preserves singular values, so its operator norm on M_n is one. At matrix level n, apply $\mathrm{id} \otimes t$ to the flip $F = \sum_{ij} E_{ij} \otimes E_{ji}$, which is unitary and has norm one. The image is $\sum_{ij} E_{ij} \otimes E_{ij} = n|\Omega\rangle\langle\Omega|$ for normalized $\Omega = n^{-1/2} \sum_i e_i \otimes e_i$, and has norm n. Hence the cb norm is at least n. Writing a matrix amplification as block transposition and using the row–column factorization gives the reverse bound n. □

17.3 What does this mean for composite quantum systems?

The transpose preserves positivity of an isolated density matrix but, as seen in Chapter 4, fails positivity on an entangled ancilla. The norm calculation measures the same obstruction quantitatively: the gap between bounded and completely bounded behavior grows with dimension. Matrix levels are therefore operational data, not decorative copies of a Banach norm.

17.4 What further contact should be proved?

Exercise 17.1 (Rows and columns are completely different)**.** *Prove that the identity maps between the underlying Hilbert spaces of R_n and C_n have cb norm $\sqrt{n}$ in both directions.*

Exercise 17.2 (Ruan's concrete identities)**.** *For $x \in M_m(E)$ and $y \in M_n(E)$ prove*

$$\|x \oplus y\| = \max\{\|x\|, \|y\|\}, \qquad \|\alpha x \beta\| \le \|\alpha\|\|x\|\|\beta\|.$$

Explain why these identities are invariant under complete isometry.

Chapter 18

Noncommutative products require Haagerup tensoring

Physical question. Which tensor product records the norm of a process formed by multiplying or composing noncommuting operator-valued inputs?

Model and assumptions. Operator spaces are concrete and finite dimensional for examples, although the definitions extend by completion. Bilinear maps are tested at all matrix levels using matrix multiplication of indices.

18.1 Why does the algebraic tensor product forget too much?

For $u = \sum_{k=1}^{r} x_k \otimes y_k \in E \otimes F$, define

$$\|u\|_h = \inf \left\{ \|[x_1 \ \cdots \ x_r]\|_{M_{1,r}(E)} \|[y_1 \ \cdots \ y_r]^{\mathsf{T}}\|_{M_{r,1}(F)} \right\}.$$

The completion is the Haagerup tensor product $E \otimes_h F$.

For a bilinear map $b : E \times F \to G$, define

$$b_n(x, y)_{ij} = \sum_k b(x_{ik}, y_{kj}), \qquad \|b\|_{\mathrm{mb}} = \sup_n \|b_n\|.$$

18.2 Which universal property fits multiplication?

Theorem 18.1 (Haagerup linearization). *A bilinear map $b : E \times F \to G$ is multiplicatively bounded exactly when its linearization extends to a completely* *bounded map*

$$\widetilde{b} : E \otimes_h F \to G.$$

Moreover $\|\widetilde{b}\|_{\mathrm{cb}} = \|b\|_{\mathrm{mb}}$.

Proof. A factorization $u = a \odot c$ with $a \in M_{1,r}(E)$ and $c \in M_{r,1}(F)$ is sent at matrix level to the corresponding product $b_n(a, c)$. Hence $\|\widetilde{b}_n(u)\| \leq \|b\|_{\mathrm{mb}}\|a\|\|c\|$; taking the infimum proves one inequality and extension by continuity. Conversely apply $\widetilde{b}_n$ to elementary matrix factorizations $x \odot y$ to obtain $\|b_n(x, y)\| \leq \|\widetilde{b}\|_{\mathrm{cb}}\|x\|\|y\|$. Taking suprema gives equality. □

18.3 Where does this tensor norm appear physically?

Multiplication in a C^*-algebra is multiplicatively bounded and therefore linearizes through $\otimes_h$. The complete isometries

$$C_n \otimes_h R_n \cong M_n, \qquad R_n \otimes_h C_n \cong S_1^n$$

distinguish operator norm from trace norm solely by reversing factors. The second norm controls optimal state discrimination, so the order-sensitive tensor geometry has an operational meaning.

18.4 What further contact should be proved?

Exercise 18.1 (Helstrom discrimination)**.** *For states ρ_0, ρ_1 with equal prior probability, prove that the best binary measurement succeeds with probability*

$$\frac{1}{2} + \frac{1}{4}\|\rho_0 - \rho_1\|_1.$$

Relate the trace norm to $R_n \otimes_h C_n$.

Exercise 18.2 (Forgetting matrix norms loses tensor information)**.** *Show that R_n and C_n have isometric underlying Banach spaces while their Haagerup products in the two orders have different norms. Explain why no symmetric completely isometric flip can identify the two products.*

Chapter 19

Channels are distinguished at the matrix level

Physical question. How well can one use distinguish two quantum processes when the input may be entangled with an auxiliary system?

Model and assumptions. Channels act between finite matrix algebras, are used once, and have equal prior probability. Arbitrary finite-dimensional ancillas and joint output measurements are allowed.

19.1 Which distance includes every composite probe?

Recall complete positivity and channels from Chapter 4. For a linear map $\Psi : M_n \to M_m$, its diamond norm is

$$\|\Psi\|_\diamond = \sup_r \|\Psi \otimes \mathrm{id}_{M_r}\|_{1\to 1}.$$

The supremum explicitly asks how distinguishability changes when an untouched ancilla is attached. Operator systems—self-adjoint unital subspaces of $\mathcal{B}(H)$—supply the matrix order that makes complete positivity intrinsic even when the domain is not a full algebra.

19.2 Why is the diamond norm operational?

Theorem 19.1 (One-use channel discrimination)**.** *For equally likely channels $\Phi_0, \Phi_1 : M_n \to M_m$, the optimal one-use success probability is*

$$p_{\text{succ}} = \frac{1}{2} + \frac{1}{4}\|\Phi_0 - \Phi_1\|_\diamond.$$

The ancillary dimension n is sufficient.

Proof. For an input state ρ on system plus ancilla, the two output states differ by $((\Phi_0 - \Phi_1) \otimes \mathrm{id})(\rho)$. The Helstrom formula from Chapter 18 gives the optimal measurement for that input. Maximizing over inputs produces the diamond norm. Convexity reduces the maximum to pure inputs, and Schmidt decomposition from Chapter 2 shows that a pure input needs Schmidt rank at most n; a larger ancilla cannot improve the value. □

19.3 What advantage can entanglement provide?

If the maximizing input is entangled, a strategy restricted to uncorrelated probes attains a smaller norm. The advantage is not additional channel use; it comes from evaluating the same map at a higher matrix level. Process discrimination experiments compare observed success frequencies with the separable benchmark and the diamond-norm prediction under calibrated noise assumptions.

19.4 What further contact should be proved?

Exercise 19.1 (Two unitary channels)**.** *For unitary channels $\rho \mapsto U_i \rho U_i^*$, express perfect one-use distinguishability in terms of the numerical range of $U_0^* U_1$.*

Exercise 19.2 (The Heisenberg adjoint has cb norm)**.** *Prove in finite dimensions that $\|\Psi\|_\diamond = \|\Psi^*\|_{\text{cb}}$ when the adjoint is equipped with operator norm matrix levels.*

Chapter 20

Bell correlations are noncommutative geometry

Physical question. How strongly can separated quantum measurements correlate without allowing communication between the laboratories?

Model and assumptions. Each party chooses one of two binary observables. The observables of different parties commute in the tensor-product model, each outcome is ± 1, and the experimental settings are statistically independent of the prepared state.

20.1 Which operator encodes the Bell test?

For self-adjoint unitaries A_0, A_1 and B_0, B_1, define the CHSH operator

$$S = A_0 \otimes (B_0 + B_1) + A_1 \otimes (B_0 - B_1).$$

Local hidden-variable assignments satisfy $|\langle S \rangle| \leq 2$. The operators need not commute within one laboratory, and that noncommutativity changes the bound.

20.2 How large can the quantum value be?

Theorem 20.1 (Tsirelson bound)**.** *Every quantum realization above satisfies $\|S\| \leq 2\sqrt{2}$, and the bound is attained by a maximally entangled pair of qubits with suitable Pauli measurements [9].*

Proof. Using $A_i^2 = B_j^2 = 1$, expand to obtain

$$S^2 = 4\,1 - [A_0, A_1] \otimes [B_0, B_1].$$

Each commutator has norm at most two, so $\|S^2\| \leq 8$ and $\|S\| \leq 2\sqrt{2}$. For the singlet state, take orthogonal Pauli directions for Alice and the normalized sum and difference directions for Bob; direct calculation gives magnitude $2\sqrt{2}$. □

20.3 What does a violation establish experimentally?

Under the stated locality and statistical assumptions, an observed value above two rules out the corresponding local hidden-variable model. It does not permit signaling, since each local marginal is independent of the remote choice. Grothendieck-type inequalities generalize the theorem: for broad families of correlation experiments, quantum advantage over classical vector assignments remains bounded by a universal constant. Loophole-controlled Bell experiments compare measured correlations with these bounds under explicitly tested locality assumptions [10].

20.4 What further contact should be proved?

Exercise 20.1 (Every pure entangled two-qubit state violates CHSH)**.** *Put a pure two-qubit state in Schmidt form and optimize planar Pauli measurements to obtain a CHSH value strictly above two whenever both Schmidt coefficients are nonzero.*

Exercise 20.2 (Grothendieck turns operators into vectors)**.** *For a real coefficient matrix (c_{ij}), formulate the classical sign optimization and the quantum vector relaxation. Show how Tsirelson's vector representation places their ratio under the real Grothendieck constant.*

Chapter 21

Infinite observables lead to subfactor index

Physical question. How can one measure the relative size of nested infinite observable algebras when ordinary vector-space dimension is meaningless?

Model and assumptions. Observable algebras act nondegenerately on Hilbert space and are closed in the weak operator topology. For the index calculation, the algebras are finite factors with faithful normalized trace.

21.1 Which closure contains physical limits?

A von Neumann algebra is a unital $*$-subalgebra $M \subseteq \mathcal{B}(H)$ equal to its bicommutant M''. Equivalently, it is weak-operator closed. Its predual selects the normal functionals, the states compatible with monotone limits of observables.

For an inclusion $N \subseteq M$ of finite factors, a trace-preserving conditional expectation $E_N : M \to N$ is the noncommutative analogue of averaging onto the smaller observable algebra. On $L^2(M)$ let e_N be the orthogonal projection onto $L^2(N)$.

21.2 Why do nested expectations produce Temperley–Lieb relations?

Theorem 21.1 (Jones projections)**.** *In the iterated basic construction for a finite-index inclusion, the Jones projections e_i satisfy*

$$e_i^2 = e_i = e_i^*, \qquad e_i e_j = e_j e_i \ (|i-j| \geq 2), \qquad e_i e_{i\pm 1} e_i = \delta^{-2} e_i,$$

where $\delta^2 = [M : N]$ is the Jones index.

Proof. Idempotence and self-adjointness hold because each e_i is an orthogonal projection. Projections separated by at least two stages act on commuting levels of the tower. For adjacent stages, use $e_N x e_N = E_N(x) e_N$ and the Markov normalization of the trace in the basic construction; applying the expectation twice contributes the scalar $[M : N]^{-1} = \delta^{-2}$. Density of the algebraic tower extends the relations to L^2. □

21.3 How does index return to quantum structure?

The parameter δ behaves like the dimension of a bimodule and composes multiplicatively. Positivity of the Temperley–Lieb trace forces the values below two to be $\delta = 2\cos(\pi/n)$, producing the discrete Jones indices $4\cos^2(\pi/n)$ [11]. The same relations yield braid-group representations; what began as relative size of observable algebras becomes data for particle exchange. The quantization theorem is deeper than the displayed relations and is used here as the bridge to Part VI.

21.4 What further contact should be proved?

Exercise 21.1 (Conditional expectation gives a Jones projection)**.** *Prove $e_N x e_N = E_N(x) e_N$ first on the dense subspace $M \subseteq L^2(M)$.*

Exercise 21.2 (Part V synthesis)**.** *Track one composite-system question through four levels: ordinary norm, matrix norm, completely positive order, and von Neumann closure. For each level identify a physical operation it controls and a counterexample showing why the preceding level is insufficient. End by deriving the braid generators from the Temperley–Lieb projections.*

Why the next framework is necessary. Operator spaces and algebras encode states, maps, and inclusions, but they do not make fusion, reassociation, exchange, defects, and gluing into first-class operations. Those processes demand monoidal and braided categories.

Part VI

Anyons, Topology, and Generalized Theories

Chapter 22

Planar exchange produces braid groups

Physical question. Why can exchanging identical particles in two dimensions retain information that disappears in three dimensions?

Model and assumptions. Particles are pointlike, identical, and constrained to a two-dimensional region. Collision configurations are excluded, motion is adiabatic, and worldlines are compared up to deformation that fixes their endpoints.

22.1 Which group records planar worldlines?

For n ordered particles in a surface Σ, remove collisions from Σ^n and quotient by permutations to obtain the configuration space of identical particles. Its fundamental group is the braid group B_n when Σ is a disk. It has generators $\sigma_1, \ldots, \sigma_{n-1}$ and relations

$$\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}, \qquad \sigma_i \sigma_j = \sigma_j \sigma_i \quad (|i-j| \geq 2).$$

Unlike the symmetric group, B_n does not impose $\sigma_i^2 = 1$.

A monoidal category has a tensor product, a unit, and coherent associativity maps. A braiding is a natural family $c_{x,y} : x \otimes y \to y \otimes x$ satisfying the two hexagon identities.

22.2 Why does categorical braiding act on many-particle states?

Theorem 22.1 (Braiding represents the braid group). *For any object x in a braided monoidal category, the maps*

$$\rho(\sigma_i) = 1_x^{\otimes(i-1)} \otimes c_{x,x} \otimes 1_x^{\otimes(n-i-1)}$$

define a representation of B_n on $x^{\otimes n}$, with associators inserted coherently.

Proof. Naturality and the hexagon identities give the Yang–Baxter equality on three adjacent tensor factors, which is the first braid relation. Maps acting on disjoint tensor factors commute, giving the second. Mac Lane coherence makes the result independent of how the iterated tensor product is parenthesized. $\square$

22.3 What physical information survives an exchange?

Adiabatic transport around a braid may act on the degenerate state space by $\rho(\beta)$. In an abelian anyon sector this action is a phase; in a nonabelian sector it can be a noncommuting matrix. Three-dimensional exchange adds a homotopy that collapses the braid to a permutation, forcing the more familiar boson/fermion alternatives under standard assumptions. Planar topology removes that collapse and therefore demands braid data.

22.4 What further contact should be proved?

Exercise 22.1 (The symmetric group is a quotient). *Prove that imposing $\sigma_i^2 = 1$ on B_n gives the symmetric group S_n. Interpret the added relation as forgetting overcrossing versus undercrossing.*

Exercise 22.2 (Temperley–Lieb generators braid). *Given the relations in Chapter 21, determine scalars a, b for which $a1 + be_i$ satisfy the braid relations.*

Chapter 23

Fusion requires tensor categories

Physical question. When anyons are combined, why can the result have several possible particle types and an internal fusion space?

Model and assumptions. There are finitely many topological charge types, fusion spaces are finite dimensional, every charge has an antiparticle, and local microscopic excitations have been quotiented out.

23.1 Which language contains fusion and exchange together?

In a semisimple rigid $\mathbb{C}$-linear monoidal category, simple objects represent anyon types. Fusion is the decomposition

$$x \otimes y \cong \bigoplus_z N_{xy}^z z, \qquad \mathrm{Hom}(z, x \otimes y)$$

is the corresponding fusion space. Rigidity supplies an antiparticle x^*. An associator compares the two orders of fusing three particles, while a braiding exchanges them. Pentagon and hexagon coherence express consistency of all composite processes.

The monodromy $M_{x,y} = c_{y,x} c_{x,y}$ is a full winding of x around y.

23.2 How is abelian monodromy classified?

Theorem 23.1 (Pointed monodromy)**.** *Suppose every simple object is invertible and the simple classes form the cyclic group $\mathbb{Z}/m$ generated by a. A scalar monodromy compatible with fusion has the form*

$$M_{a^r,a^s} = \exp\left(\frac{2\pi i k r s}{m}\right)$$

for some $k \in \mathbb{Z}/m$.

Proof. The hexagon identities make monodromy multiplicative in either variable, so M is a bicharacter on $\mathbb{Z}/m \times \mathbb{Z}/m$. It is determined by $\zeta = M_{a,a}$. Since $a^m \cong 1$, multiplicativity gives $\zeta^m = 1$; write $\zeta = e^{2\pi i k/m}$. Bilinearity then gives the formula. □

23.3 What phase results for one-third anyons?

For $m = 3$ and $k = 1$, winding a once around a multiplies the state by $e^{2\pi i/3}$. In an interferometer, changing the number of enclosed anyons therefore advances the interference phase by $2\pi/3$ in the ideal abelian model. Real devices also include Coulomb coupling, edge reconstruction, and decoherence; the categorical prediction concerns the topological contribution after those effects are controlled [12].

23.4 What further contact should be proved?

Exercise 23.1 (A three-cocycle controls reassociation)**.** *For G-graded vector spaces, multiply the associator on homogeneous degrees g, h, k by $\omega(g, h, k) \in U(1)$. Prove that the pentagon is exactly the 3-cocycle equation and that changing ω by a coboundary gives a monoidally equivalent category.*

Exercise 23.2 (Fusion dimensions)**.** *If positive numbers d_x satisfy $d_x d_y = \sum_z N_{xy}^z d_z$, prove that d is a positive character of the fusion ring. Compute it for the Fibonacci rule $\tau \otimes \tau = 1 \oplus \tau$.*

Chapter 24

Chern–Simons theory turns braids into amplitudes

Physical question. Which field theory assigns an amplitude to a braid using only topology rather than a spacetime metric?

Model and assumptions. Spacetime is a closed oriented three-manifold, the gauge group is compact, and the level is chosen so the exponentiated action is gauge invariant. Path-integral statements are interpreted formally unless a rigorous TQFT construction is specified.

24.1 Which action sees knots but not distances?

For a connection A on a principal G-bundle, the Chern–Simons action is

$$S_{\mathrm{CS}}(A) = \frac{k}{4\pi} \int_M \mathrm{tr}\left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right).$$

It uses orientation and a trace pairing but no metric. Variation gives $F_A = 0$, so classical solutions are flat connections whose holonomy may still be globally nontrivial.

A labeled closed worldline K defines a Wilson loop $W_R(K) = \mathrm{tr}_R \, \mathrm{Hol}_A(K)$.

24.2 Why must the level be quantized?

Theorem 24.1 (Large-gauge invariance)**.** *With the standard normalization for compact simple simply connected G, a gauge transformation $g : M \to G$*

changes the Chern–Simons action by $2\pi k \deg(g)$. Hence $e^{iS_{\mathrm{CS}}}$ is gauge invariant for all gauge transformations exactly when $k \in \mathbb{Z}$.

Proof. Substitute $A^g = g^{-1}Ag + g^{-1}dg$ into the action. Cyclicity of trace separates an exact transgression term and

$$\frac{k}{12\pi} \int_M \mathrm{tr}(g^{-1}dg)^3.$$

The exact term integrates to zero on closed M. Under the standard trace normalization, the remaining integral is $2\pi k$ times the integer winding number of g. The exponential is therefore invariant for all winding numbers precisely at integral level. □

24.3 How do Wilson loops recover anyon data?

In Chern–Simons TQFT, exchanging worldlines evaluates the braid morphism of a related modular tensor category, and closing a braid evaluates its categorical trace. For suitable $SU(2)$ level, Wilson-loop amplitudes produce Jones-type link invariants. Thus fusion and braiding are not appended to the field theory: they are the invariant data assigned to punctures and worldlines. The formal path integral motivates this relation; rigorous Reshetikhin–Turaev constructions implement it categorically. [13, 14]

24.4 What further contact should be proved?

Exercise 24.1 (Flat does not mean trivial)**.** *Construct flat $U(1)$ connections with distinct holonomy on a three-torus and compare with Chapter 13.*

Exercise 24.2 (Wilson loops respect braid closure)**.** *Assuming the ribbon-category evaluation rules, prove that isotopic closed braids have equal amplitudes and identify where framing enters.*

Chapter 25

Functorial theories stabilize observables

Physical question. What common structure turns fusion, gluing, bundles, and Fredholm operators into deformation-stable numerical observables?

Model and assumptions. Categories are additive where Grothendieck completion is used, tensor products distribute over direct sums, and generalized cohomology is evaluated on finite CW complexes unless stated otherwise.

25.1 How are extended processes composed?

A strong monoidal functor $F : \mathcal{C} \to \mathcal{D}$ comes with coherent isomorphisms $F(x) \otimes F(y) \cong F(x \otimes y)$ and preserves the tensor unit. A three-dimensional TQFT is a symmetric monoidal functor from a bordism category to vector spaces: disjoint union becomes tensor product and gluing becomes composition.

For an additive monoidal category, its Grothendieck group is generated by objects with $[x \oplus y] = [x] + [y]$ and multiplication $[x][y] = [x \otimes y]$. This is decategorification: fusion spaces and morphisms are forgotten, while stable addition and multiplication remain.

25.2 What survives a monoidal passage?

Theorem 25.1 (Monoidal decategorification)**.** *An additive strong monoidal functor induces a ring homomorphism*

$$K_0(F) : K_0(\mathcal{C}) \longrightarrow K_0(\mathcal{D}).$$

If the categories are rigid and traces are preserved, categorical dimensions and traces of closed processes are preserved as well.

Proof. Additivity gives $[F(x \oplus y)] = [F(x)] + [F(y)]$, so the assignment $[x] \mapsto [F(x)]$ respects the defining Grothendieck relation. The monoidal comparison gives $[F(x \otimes y)] = [F(x) \otimes F(y)]$, hence multiplication is preserved. Strong monoidal functors carry evaluation and coevaluation maps to duality maps, so the composite defining categorical trace maps to the corresponding composite in $\mathcal{D}$. □

25.3 How do generalized theories unify the earlier invariants?

A reduced generalized cohomology theory is homotopy invariant, sends cofiber sequences to long exact sequences, and has suspension isomorphisms. Complex K-theory is two-periodic and admits equivalent bundle, projection, unitary, and Fredholm models. Those models connect the occupied Bloch bundle of Chapter 15 to the index of Chapter 16.

The recurring pattern is now explicit:

$$\begin{aligned}&\text{physical structure} \longrightarrow \text{functorial stable class}\\ &\longrightarrow \text{pairing or trace} \longrightarrow \text{observable.}\end{aligned}$$

Aharonov–Bohm phase, Hall conductance, Fredholm index, and anyon amplitude use different categories, but each is protected because the middle class is unchanged by the allowed deformation.

25.4 What further contact should be proved?

Exercise 25.1 (Cohomology preserves supertraces)**.** *Prove that cohomology is symmetric monoidal on bounded complexes of finite-dimensional complex vector spaces and deduce the Hopf trace identity.*

Exercise 25.2 (Part VI synthesis)**.** *Start with the pointed $\mathbb{Z}/3$ braided category of Chapter 23. Construct its braid representations, close a braid using categorical trace, interpret the result as a Wilson-loop amplitude, and decategorify its fusion rules. Identify which data are lost at each passage and verify that the $2\pi/3$ monodromy phase remains.*

Exercise 25.3 (Final prerequisite audit)**.** *For one invariant from each part, list every mathematical structure on which its definition depends. Order those prerequisites independently of the book's narrative and verify that every dependency appears before its first use. If an assumption belongs to a physical model rather than a theorem, mark it explicitly.*

> **Closing perspective.** Superposition demanded linear spaces; invariant transformations demanded representations; local comparison demanded geometry; robust global response demanded topology; matrix-level composition demanded noncommutative analysis; and fusion with exchange demanded categories. At each stage, the failure of the preceding framework selected the next coordinate-free structure.

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